tf.expand_dims Stay organized with collections Save and categorize content based on your preferences.

https://www.tensorflow.org/api_docs/python/tf/expand_dims

> *Returns a tensor with a length 1 axis inserted at index axis.*

Function Signatures

```
tf.expand_dims( input, axis, name=None )
```

Code Examples

```
tf.expand_dims(
input, axis, name=None
)
```

```
tf.expand_dims(
input, axis, name=None
)
```

```
image = tf.zeros([10,10,3])
```

```
image = tf.zeros([10,10,3])
```

```
tf.expand_dims(image, axis=0).shape.as_list()
[1, 10, 10, 3]
```

```
tf.expand_dims(image, axis=0).shape.as_list()
```

```
[1, 10, 10, 3]
```

```
tf.expand_dims(image, axis=1).shape.as_list()
[10, 1, 10, 3]
```

Documentation

Returns a tensor with a length 1 axis inserted at index axis.

Used in the notebooks

• Extension types

- Import a JAX model using JAX2TF
- Migrate `tf.feature_column`s to Keras preprocessing layers
- Understanding masking & padding
- Working with RNNs
- Integrated gradients
- Playing CartPole with the Actor-Critic method
- Generate music with an RNN
- DeepDream
- pix2pix: Image-to-image translation with a conditional GAN

Given a tensor input, this operation inserts a dimension of length 1 at the dimension index axis of input's shape. The dimension index follows Python indexing rules: It's zero-based, a negative index it is counted backward from the end.

This operation is useful to:

- Add an outer "batch" dimension to a single element.
- Align axes for broadcasting.
- To add an inner vector length axis to a tensor of scalars.

For example:

If you have a single image of shape [height, width, channels]:

You can add an outer batch axis by passing axis=0:

The new axis location matches Python list.insert(axis, 1):

Following standard Python indexing rules, a negative axis counts from the end so axis=-1 adds an inner most dimension:

This operation requires that axis is a valid index for input.shape, following Python indexing rules:

This operation is related to:

- tf.squeeze, which removes dimensions of size 1.
- tf.reshape, which provides more flexible reshaping capability.
- tf.sparse.expand_dims, which provides this functionality for

tf.SparseTensor

Returns

Raises